

Lab 03 - Vigenere Cipher

Intro

In this lab you will solve a Vigenere cipher which is polyalphabetic. This cipher was labeled as indecipherable for over three hundred years until Kasiski published a method for decrypting the cipher. You will use the Kasiski method in your lab as it is easier than calculating the full index of coincidence (I.C.).

Part 01

1. [Login to iselab](#) and power on your machine “cpre3310_[netid]_Ubuntu_22.04_desktop”
2. Change into your homework directory. Scp the skeleton code. You will get a ciphertext.txt file and skeleton code. It should create a new lab03 directory with the skeleton code inside. The credentials are repo/repo.

```
cpre3310@cpre3310:~/homework$ scp -r repo@repo.homework.3310.com:/home/repo/lab03/ .
```

3. Don't forget to give yourself write permissions inside the lab03 directory.

```
cd lab03
chmod 750 part01
cd part01
chmod 640 *
```

4. Edit the skeleton code `part01_skel.py`. Most of the hard work has been done for you.
 - a. The program conducts a trigram analysis using the sliding window on the ciphertext.txt file.
 - b. It calculates the starting indices for the most common trigrams.
 - c. It also calculates the differences between the starting indices for the most common trigrams.
 - d. Look for the #TODO: indicator in the code. You will need to use the findGCD function to determine the common factors of the differences.
 - i. The common factor(s) are the possible candidates for the keyword length.
5. Once you have the findGCD function completed, you should run the program using `python3 part01_skel.py ciphertext.txt`.

- Below is EXAMPLE ciphertext and the completed program. Please use your own ciphertext.txt and code for your lab report.
- You will first be shown the ciphertext has been loaded.

```
cpre3310@cpre3310:~/homework/part01$ python3 part01_skel.py ciphertext.txt

Ciphertext Loaded:
JFL LXIPASIYRMOYRHCBFQCAQXSENLSL FVHGRGHLNLHP CZFQXBAZXJLHWZLFBWZNRHVACAHXNOZBDBKH WZLFBWZNRHVLXIBAMS YFCOUQMCFB
DIUQNFZGJBKLNKGBW HHOAHTRRAORASABOSSCHCBENSZRBSTYN OUGCSJULCTNBGPTWSKGXDYBCSJGHCBLXICRKS LACOYTNHARMTVECSYZRBHG
RCUVMCUGTBVJRHSBXYZYRYLVCAPTQHZCASHQCCAURGMHABPGDFLJJFLUXIZRJQYBBGAUNOSYNMAUNDHVH VACVLJJZSFBHHECWUTCCIYRGAR
AIWONHARAULGJBVGQSYGAIJXACBAMHVGQWZFRRLGQWZVBOTVBH HXNWNONESUGMCURJBFGQWUTWC IHCMVHFWSYRHZINFFVVDVECOUGCVHGHCBY
RJLVLOUGKSSVNJLGQWZVBVHCYSUVWUOB FQVHURAUJBTNWULGDDHSCSYLXIUBCOTNWOARAAPAJHVELMIRARFAJAPPB AVQNZZHB DLPCJLURQSR
BWNUCSKBWAVGXFHGYWJBBCBGQPVHWRBARHZGFCGRACZVGOUQWCRBSCRWOAGNAWGRBARAQLCC IUVCCUROCBECVYRNQVZNWUGQWZVBCUROCB
CVYRNKLCFCPVHWRVAXZFZYWJNYDYBJQOVWUVINF SNWRHVAIUUCHDBFSYRXBOVVKLF CFPVHWRZNWHH ZXPBJOASXIYMN FVSRJLNV OJURBL LXITR
JBSVTSHEXPVGCANACIBCQFOXFNPHLEWSAVLCY TJBPFPVBGQSDNBPSNRNPAPXBF C0ZRLCUQTS LCHCBEQSHQMCD AUZCQGWTTXCKPXJLEJZYV
PVAYRGARWHORCSYZRBHGXFZNWUUSRZAEJHPBW IUUCDECAHAYOYGVOJURBLHWRL EWSHGQWAFJVF C NFHYUCFPXAINCQONBGPFVWYPXDYBLSZF
XFJBWHYBUZLQOISYHOYZXFLQESYLC CBTOPBGXIAFRRLVCGSVEWUTQITNWHPFBI LSUSZUBYP AQOPEKZVB MUYBFBMBAHORLMTBAUZYXC RENSR
RYUBFMVHFUOUGCCORUDIHC DHLJHARWHPBWHORBSYVNGONMFBOKSYFTWUJNGWBCHLQC VLZNOZLKIAGQSZRJFLANKAUNMSBX YOHOUFFSHGKOKO
ASHGQSCRAMAURBNINFFUJFKGXGWB CWNMHVJJWAGRZORVCCRM CULXI IROCYRRQVHURGRACOVVLLRAUBCGAHYWKLTB VJCVLLLOUGKIPYMOUL
CVPAPZPXNHONCMLGWCUBCMLGWCASXFHOXIASXFALHSHEBGVVCMEXAUNTBGDFLVBHONCFPTQHVANDVFBW IYNTBGDFLSXIYLXIYCXWUGXTCV
NKPOXBAXWCDGQSARLVZGDTMNRFBDFLSACTGQSMHC IYRCCVERUOGAWNUCQFOXFNFM CUGOSLYYOPARRVQXBAQHONCONNRBQBHSRCALT XZPF
CSUHWRL EBBHAMHONCHLEVVUNCCYVBCBGCVL ENWAPJBAONFLNB CURMKPGQWAPJBAONPHEPOPANRDVCVPGMCLFWHMRNZWCMVSASTBAGLBATLN
AOUQRHHOBCSHCSSLFWSYWCAFCWRESYHWHPHYCBNASKRJRJNWMVBHVC RHTNHPLJRHO GQSZRFSHCXBZVMCUGTBVJBJZCNQAINVPUS SBLOR
MOACJFRVWUSBCQLQJFHAMUSRWVHINBYRNLJQMT RFVFXSZVCKHACALGQSYRBGVZDQOGNZSZNXBFCGANAHHGCVLONUPAWWUTCVL ENKHFJKHE
JTLJHSHEBTYBVBVJWIJYNOYJJFAUNKOBUS AURBNNUZAU RGLNFFGOWUTRGNBWSQHBNBWSAUNFLJNFLDFD FCECYFQSYR CVLENBVOXRF XWSDJ
QCZGJFARMWAVCKHFCVLZJQOVWSZVMCUGDBKRAGANWRKROSUFNBLGFCYXLTCTDHL EWSDCXKLEOISUXCRRMWUGXSCRAMAURBNGAIZGNRABAIUV
COSYCVLLBOFVCUVGBAHECOURFCYQNFVSRBARUZPTNB JRCVLARHZNFOSYYSVCUSHFJHOENOAAXHQHBAUNCURBCUGQSVGQSYFRRLQNPQNRVH
CTHGNWUNVWJEXGLPXBKRGHLEVWUNCWVAMWKACGLRCLVJJFPJJGIBABHSCSYVW HORAIPABUYRFI WGQSYRBHHEEWUTQWKVWUMEXAUNVRFCVLJ
QOAUDBARAYPYUSYFYOAEXZTNLVPANGIHRZKVWOBGXAHGNRMNLHVERSZZXGABOIZJNFLEXIUQNRBCYIAVWQHZYGM BACYQNFSLMWZCXGHYKIYA
NRPAKMSNB SYFLOUFXALBOIZJNFLXNDANUWCRCDBAYSBJRPAPPVQRSZGQSKVBDVFJZBARHZEJB UVV PVANWRKNHKLJNFLGQOAPUCZRC CNBRBNB
DHMBASCRAPBGCVLENKHF XBLZJB DUXHHHPVAHBHVS RUOGCCZGXFTGQSDVASVSVLPJAW FCCZZJGOGQCRV SANUGBPTSYFRBABSIXQSAHABLQ
RHHEXIUQQSIEIXINUCIZOJQRSACTGQSIERBR CJIZRQWZAJALVBQVAWCYWXVUPXUBAMVHAGVABOYNQM VHAIUOXFUFXB
```

- It then displays the five most common trigrams, the count for each, the position the trigram is found in, the differences between the starting indices, and then it calculates the common factor for each. Take a screenshot of the five most common trigrams for your lab report. Finally, you are prompted to provide the key length which is the common factor(s).

```
Most Common Trigrams:
Trigram: CVL
Count: 14
Positions: [300, 1144, 1280, 1544, 1812, 1824, 1928, 1960, 2056, 2096, 2200, 2264, 2496, 2544]
Differences: [844, 136, 264, 268, 12, 104, 32, 96, 40, 104, 64, 232, 48]
GCD: 4

Trigram: GQS
Count: 14
Positions: [339, 803, 1155, 1191, 1415, 1439, 1687, 1783, 2139, 2143, 2235, 2431, 2535, 2619]
Differences: [464, 352, 36, 224, 24, 248, 96, 356, 4, 92, 196, 104, 84]
GCD: 4

Trigram: GQW
Count: 9
Positions: [355, 363, 391, 447, 635, 847, 931, 1567, 1887]
Differences: [8, 28, 56, 188, 212, 84, 636, 320]
GCD: 4

Trigram: AUN
Count: 8
Positions: [282, 290, 1166, 1350, 1862, 1906, 2130, 2258]
Differences: [8, 876, 184, 512, 44, 224, 128]
GCD: 4

Trigram: LXI
Count: 7
Positions: [3, 91, 187, 487, 751, 1239, 1391]
Differences: [88, 96, 300, 264, 488, 152]
GCD: 4

What is the likely key length?
```

- d. Why do we analyze trigrams (three-character sequences) when attempting to break a Vigenère cipher? What information do the differences between their positions provide?
- e. How does calculating the greatest common divisor (GCD) of position differences help in determining the key length?
- f. In this example, you are lucky because they all factored to 4. You may have multiple answers, but you should be able to reduce the factor to one to try.
- g. Based upon your findings above, what is your keyword length? You need to record this in your lab report and justify your response. After entering the keyword length, the ciphertext is broken into that many monoalphabetic shift-by-n ciphers and the character frequencies calculated. These are displayed on the screen.
- h. The top 3 characters in each monoalphabetic shift-by-n cipher are provided with the suggestion to set E to that character. Why is that assumption made? How does it help us with multiple shift-by-n ciphers?
- i. Take a screenshot for *each* shift-by-n cipher just like the screenshots below. If your keyword is 4 characters, you will need 4 screenshots. If your keyword is 5 characters, you will need 5 screenshots.

Remember, the output below is just an EXAMPLE. Your output will be different.

```
Analyzing Cipher 0:
Frequency for Cipher 0:
C: 86
N: 74
X: 68
W: 53
R: 48
B: 44
A: 44
J: 43
Q: 43
M: 25
F: 16
U: 16
L: 15
H: 14
D: 14
V: 14
Y: 11
O: 10
K: 9
T: 7
P: 7
E: 6
S: 2
G: 2
Corresponding alphabets:
C: Shifted Alphabet (setting E to C): YZABCDEFGHIJKLMNPOQRSTUVWXYZ
N: Shifted Alphabet (setting E to N): JKLMNOPQRSTUVWXYZABCDEFGHI
X: Shifted Alphabet (setting E to X): TUVWXYZABCDEFGHIJKLMNPOQRS
```

```
Analyzing Cipher 1:
Frequency for Cipher 1:
S: 80
C: 70
H: 56
B: 48
W: 45
O: 44
I: 43
F: 41
V: 33
G: 32
R: 28
Z: 20
A: 20
Q: 18
M: 16
U: 14
D: 13
K: 13
P: 12
T: 9
Y: 7
J: 6
X: 2
L: 1
Corresponding alphabets:
S: Shifted Alphabet (setting E to S): OPQRSTUVWXYZABCDEFGHIJKLMN
C: Shifted Alphabet (setting E to C): YZABCDEFGHIJKLMNOPQRSTUVWX
H: Shifted Alphabet (setting E to H): DEFGHIJKLMNOPQRSTUVWXYZABC
```

```
Analyzing Cipher 2:
Frequency for Cipher 2:
L: 82
A: 68
U: 60
V: 53
Y: 52
H: 50
Z: 48
P: 36
O: 29
S: 28
B: 26
K: 18
F: 16
T: 16
I: 13
N: 13
J: 12
C: 10
M: 10
D: 10
W: 8
R: 6
Q: 3
G: 2
E: 1
Corresponding alphabets:
L: Shifted Alphabet (setting E to L): HIJKLMNOPQRSTUVWXYZABCDEFG
A: Shifted Alphabet (setting E to A): WXYZABCDEFGHIJKLMNOPQRSTUV
U: Shifted Alphabet (setting E to U): QRSTUVWXYZABCDEFGHIJKLMN
```

```

Analyzing Cipher 3:
Frequency for Cipher 3:
G: 81
R: 74
B: 50
V: 48
N: 45
E: 43
F: 39
A: 36
H: 32
U: 27
Q: 26
L: 21
J: 20
Y: 19
C: 17
P: 16
T: 16
Z: 15
S: 15
O: 13
X: 9
I: 6
M: 1
W: 1
Corresponding alphabets:
G: Shifted Alphabet (setting E to G): CDEFGHIJKLMNOPQRSTUVWXYZAB
R: Shifted Alphabet (setting E to R): NOPQRSTUVWXYZABCDEFGHIJKLM
B: Shifted Alphabet (setting E to B): XYZABCDEFGHIJKLMNPOQRSTUVWXYZ
Enter the keyword you think will decrypt the ciphertext: █

```

- Notice you are now being asked to provide the keyword (in this case the four letter keyword) that was used to encrypt the message. A first trial would be YOHC. This would be made by choosing the most commonly found character in each shift-by-n monoalphabetic cipher and then setting E to that value.

```

Enter the keyword you think will decrypt the ciphertext: YOHC

Decrypted Plaintext:
LREJZUIYUURPOARPJ0UDSOTOZEXLNTLJHHAETSAJXPANELY0ZNTXZVEFYLEDISLJ00YEMAVPASZFNDFYLEDISLJ00JZUUY0ERDEAN000YZFUN0PSELNDJPSDZYTHFCMTMHPCEZS
ELAJ0UCPESPDERRPANEEECNOMLDSIRYEDEZPRZECEJ0UJZUVPMEEYEARPPPTPOFOCEERTNAETONT00NEVNOHTTLZZKSWTKETEMIRSTACEA0E0TSTFFCNIEFREHLRESZUSPLCRZDS
TSPALWPTYSPPATYTOYEHEHLLDDTACEINRE0BWTSTPCUPMPTTPCGEELNOESEREUCVC0UY0T0ESISDTDEESIST0AMTDTAVPIHLGENE00NPLNYESINRY0BFEY0FHLWTTSGPRYTXPOCEAN
EEHAEJ0UWTVETNANEMELTPVEESISTDHAAAEITYGHZCOFWDTSLNMLYGEFFPAQEERJZUNZEAMLYATPCMIYLT0CNYBPCDYLYMINDMOOPLSFDPENEVESTCLPDIGSEEDZYMOEZRAEAI0ZDOU
SB0FYDUYT7SEHOZPCOSTIAN0QIVPDEVYATEPMPEINTPCCEAEUNTEONP00UCEHRPPCOXPINESISTDONP00UCEHRPPWEDEBOFYD0YZLYXAICLAPRZLCHTYG0GPRLLYDATCUNTETWZHERP2
NHTXWEDEBOFYD0SLYTAXZINLATQZURKPROQTVELXACSTNEJZUMPLNLTVEACZBOEYOTLC0BZECYMRGNJBECYETTNRRLNIDXBUESEWLDLPPDIYRJUDEASPNO0VEEAJ0UCSEA000WYWC
SESIMRZODNZVECLLRHTRTWSTPYTHPEERTNAEZRSLYINQTLTCLTIZYUNTEPACEMAYAAEXACSTNEFYDECYEAESITDLHYAPRAWOYNZMBLECHLDSIDXIRNZPRZNE0SDZRCZYTRZWL0EQUUL
WJARXZRE0GERJEDURSBU0EZUTD0TDETESLTGINRSUMLYTI0DDUEQWESSDKIYSAICML0Z0GRZHNFCZTHPNYBZCGS0W0K0CPESPTKNZHY0FHANEE0HPWPBFEP0AJLTTPTTYIZYTHPDERTPSHLORUK
MERDVINHPSPZETE0EHXPASJ0MUTESESPLREY0WTSPLYLZZKHFXANDHEAEMADMCEAESEVPCYTSNNGGPRYSLRDEZSPZEIHL0T0HLITETLHPX0VP00NJZUBPQ0RPTC0FWDZPC0HTXHEJTMNZ0
STFAIDJVN0HEHEJNANEMUI0ANJEHIYRLIVPTHLEYEY0NZEY0Y0TZRAMZUT0ZRTJJEACDS0TESFCZMTSPFUEFRETDTHLERIRSTOYPP0DDIBWP0FUEFREQZURJZURAZINEZ0FVTPWIOZ0N
TVY0W0E0SETPNHSEFFFLYDYZFREQ0MESEFFEURPE00CTGHECIGSECYMZRG00NEQE0EWAAYTD00ZNT0ZTHLEAGLTJNFDLTPEMERZLIDEENFYDEC0DTAY0THLETECXINLE0RTD0UEEHECPTI
NLNTMPRELDONPOWIESITNLNTMPBACRAIYPDWEHIE00EDYTFPPLPTEY0QCEMZCSEZCFELCAN0TTAM0DLFEELJHILWY0TDE0PPGERFYTIWJ0ULCEDPLDCLY0F0T0ATTMLJBEHTTHSES0F
HEAAZNS00NEVNOHDUSAPCTGPHINWELZNAT0ATLARKTYGLZECE0LRAY0GLPYHAGPNRPPSEHSYMPHY0ZESTEWAYEMEESERPDS0X0F0CHEPLXPJ0DESTLCTAEEHEMPGIYYINREHECPWADU
WACLFEHJ0ACDFRZXNOHYUCPARHLRTSPWHZ0WETSTNGLWLTSTSEGP0RYESINRTSGZY0EJ0DGTZY0ETSPREHPRED0FRVTGORDSERPEHECPNOMZ0VY0EWS0SE0LRT0ITTEW0ADEH0EXLCHT0YEST0C
NEFNDPCSTLYDDPQ0NDPN0E0HORVNO0MAFTECY0EWA0Z0W0C0ULS0Z0K0P0INE0Z0V0CYT0STNGECUSEPDTZCUNTEALWEHEJDAYTEGOEDMACEANPHOR0PRO0TNTPWLIRPNCPEHEYITSLHALWAE0AWE
DLTHCPATY0ZTJ0D0TSPONPD0NE0E0SER0TDE0PCIO0P0D0FE0FAEPINLXIC0CZ0SEN0ZNDPITECXINLEI0Y0IDY0ESE0PEHEHLRIHLSBZCNA0E0ERTY0THPCUIYDGRPHUP0ESERPDTACGINRSIDTY0FC
ZMTSPHKDEHEHSATSFNTPCKI0W0ERDAATCZMLNHIY0PSBFTLDYAU0E0ZMA0EPD0FLNTOCTESX0Z0T0Q0USHPRE0CZUN0PD0UAAUTTYCAXAS0FZCOR0PRLJ0ISAZS0AWMURY0PDY0MYLLDER0NAND0ZME0Z
J0SHPRE0VPPTLWIVPE0WZCKLZLDIYR000TESE0EDT0P0D0LLUYT0TSC0LNTRHTLYDDLJWEHPRE0SATN0S0PE0GZTNGZFTFZCEVP0CBUE0HECPWAD0ZNE0X0LNS0ZTAFRHTFD0T0GTGHE0E0SE0ZRMSE0
WTCE0QE0HENLMPDE0SXL0SHES0SPXETLWSUNV0ERTNTZUUNV0SETFCNE0TTACZUN0SEB0CZUG0EUSMLK0C0MESEB0CTNKALUSP0SISY0METDC0Y0R0ZHNNZN0NZCY0FC0Y0DARL0Y0FCUNM0ZRN
0ZN
Do you want to try another keyword? (yes/no): █

```

- If you look at the output, do you see words and if so, what position (shift-by-n monoalphabetic cipher) is correct and which may need to move to the next most frequency letter. I will enter `yes` and then try the keyword `JOHC`


```

Enter the keyword you think will decrypt the ciphertext: JOHC

Decrypted Plaintext:
AREJJOITYJURPDARPOUDHOTOEXLCTLJWHAEISAJEXANTLYOONTXOVEFNLEDISLYDOYTMAVEASZUNDFNLEDISLYDOJOUUYDERDTANODOYUUNORSEANDJESDZNTHTFRMPIMHPRETZ
ELAYOUCCEESPERREANETECSCOMLSSIRNEDEOPRTZECEYOUJOUVPBEEYTARRETTPDFOCTERXINAEONTONEKNOHITLZOKSWIKETMTIRHTSAREAOOTSTISFFRNTUEUHARESOUSPACRZSS
TSEALWEYSEPATNTOTYTHEALLDSTACTINRTOBWISTRUPMETTPRGEANOHEHERUCVROUYDOTOEHISDIDEEHISTSAMTSTAVEIHLVENEDONPANYEHINRNOBFTYOFNLWITSGERYTMOCTAN
ETHAEYOUWIVETCANEBELTEVEEHISTSAAPENTNGHZWCOFLDTSANMLNGEUPAQATERJOUNZTAMLNATPRMIYATOCYBPRDYAMINSMOOLESFSPENTVESICLPSIGSTEDZNMEOERAEPICZSOUE
HBOFNDUYITSEWOZPROSTXANOFIVPSEVPNATEEMPEINTPRCEATUNTNTONPFOUCHTRPECOXEINEHISTSONPFOUCHTRPEWEDTBOFNDOYOLYXPICLPPRZACHTNGOGERLLNDATRUNTTTWZERPO
NHTMWEDTBOFNDSLNTAXONINAAOTOOURKEROOIVELMACSINEJUMPANLTKACOBDOENOTLROBZTCYMORGNBYECNETTCORRANIDMBUEHEWLSBLPEDIYGDUTASPCONOKEEAYOUCHEAODOWYLO
SEHIMROODNOVECALRTGHTWISTPNTHPTERXINAEORSLNINQILTCATIZNUNTTACTMAYPAREMACSINEFNEDECNEAEHITDAHYAERAWLOYNOMBLTCHLSSIDMIRNOPRZCESDORCZNRZLLEOFUM
WYARXOREOVERJTOURHBUEOUTDIDETTSLTVINRHUMLNTIDSUEQLESSSKIYHAICBLOZDGRZWNFZRTHPCYBZRGSWOOKCEESPKNZWOYFMANETOHLBPFTPAJATTPTNTIZNTHPSERTESHLDROM
BERDKINHESPTZTEOTHEXEAJBUTEHESPAREYEWTSYELZOKHFMANDWEAEABDMREAHEVPRYTSINGGERYSARDEOSPZTIHLDTOHAITEILHPMOVPOONJOUBPFORPICOFLDZPROHTMHEJIMNZT
STFPIDJKNHTHEJCANEBIWDANJTHIYGLIVETHLTYEENONZTYEENOTQORAMOUTQORTJYEACSSOTTSCOMTSEFUEURETSTHLTRIRHTOYEPDOSIBWFEUEUREQOOURJOURAONEOFVTEWIOON
TVNWEHETPCHEUFFLNDYZUREQROMEHEFFTURPTOOCIGHERIGSTCYMORGDDONEFEFEWPAIYIDOOONTOOTHLTAGLINJFSTLPTMEROLDITENFNDECSTAYDTHLTTECMINLTORTSOUTHECEIT
NANTHERELSONPDWIEHITNANTHEBACGAIYEDWTTHIEDOEDNTFPELPTTYOQREMZRSEZRFELRANOITAMSOLFTELJWLWNOTDTPPPVERFNTIWOULREDPADCLNYOFSTOAITMLYBEHITHEHESP
WEAANSTDONEKNOHSUSAECTGEHINLELZCATPDATAARKNGLZTCEOARAYDGLPNHAGENRPESEHHYMPWHYOESTTWAYTMEEHERPSSOXUCHEELLXBJUDTSLTRTAETHEMEGIYNINRTHCEWADA
WACAFEHYEAACFRZMNOHNUCWEARHARTSEWHZLETSINGLLTISISEGERYEHINRISGZNEJFSTGZNETSEREHEREDURVTORDHERPTHECENOMODYVNEWHHSEARTPDITTTWADTHEXACTHNESTDO
NEUNDPRSTLNDPFFENDENEWORVCOMAUTECNEWAOWECFULSOOKPDINEOEVPRTYSINGERUSEEDTZRUNTTALWTHEJSAYTGOESMACTANPWOROEROINTPLLIRENCPTHEYITSLWALWPEOLEA
DATHCEATYOTJFSTTSEONPSONEHEOHERIDEOECIOEDOTTFEEINLMICCOSENONDPXTECMINLTIOYDIDYTSPTHEHARISHASBZRNAOTERTNTHPRUIYSGRPWUPEHERPSTACVINRHIDTNGFC
OMTSEHKDTHEHHTASUNTPRKIWLERDPATCOLMLCHYESBFLDTNAUEOMAEFDLCTOCIESXOSTZFUSHHERECOUNOEDUAPUTTNCAXPFSZROROERLJDISAOSAWBURYEDIYBYLLSERDCANDOMEZF
USHEREVEPTLLLVPTOWZRKLZADIYGBOOIESEHEDTSPODALUYITS CANNTHGHTLNDOLYMEHEREHATNLOSPOTGZINGZUTTFZREVPRBUETHECEWADONEXANWSOTAFGHTFSTOIGHETOSOEORMEHE
WTREOQTHENAMPDTOSXASHEHOSPMETLLSUNKERDINTZJUNVHETFRNEOITACOUNOHEBCOUGSTUSMACKOROMEHEBCINKKAUSPHISYAMETSOCYONORUOHNNONNZRYOFRSOYSARLHYOFRUNMORN
DON

Do you want to try another keyword? (yes/no):

```

8. This is closer and I think the last letter is wrong. I think the keyword is JOHN and there is an N in cipher 3, so I am going to try it. What strategies can you use to refine your keyword guesses if the plaintext is not immediately recognizable?

```

Enter the keyword you think will decrypt the ciphertext: JOHNN

Decrypted Plaintext:
AREYOUINJUREDAREYOUSHOTDOEXACTLYWHATISAYEXACTLYDONTMOVEUNLESSISAYDONTMAKEASOUNDUNLESSISAYDOYOUUNDERSTANDDOYOUUNDERSTANDYESDONTHURTMEMHERETOH
ELPYOUREESESERGEANTTECHCOMASSIGNEDTOPROTECTYOUYOUVEBEENTARGETTEDFORTERMINATIONIDONTKNOWITLOOKSLIKEITMIGHTSPREADTOTHSFURNITUREWAREHOUSEACROSS
THEALLEYTHEPAINTONTHEWALLSSTARTINGTOBLISTERUPBETTERGETANOTHERTRUCKROUNDTOTHISISIDETHISISAMISTAKEHAVENTDONEANYTHINGNOBUTYOUWILLITSVERYIMPORTANT
THATYOUHAVEICANTBELIEVETHISISHAPPENINGHOWCOULDTHANMANGETUPAFTERYOUNOTAMANATERMINATORCYBERDYNAMICSMODEL SUSPECTVEHICLESIGHTEDONMOTORATPICOOUT
HBOUNDUNITSTWOZEROSIXANDFIVESEVENATTEMPTINTERCEPTUNITONEFOURTHREECOMETHISISONEFOURTHREEWESTBOUNDONOLYMPICAPPROACHINGOVERLANDAIRUNITTWOWEREO
NHIMWESTBOUND SANTAMONICAATFOURZEROFIVEAMACHINEYOU MEANLIKEAROBOTNOTAROBOTCYBORG CYBERNETICORGANISMBUTHEWASBLEEDINGJUSTASECONDKEEPYOURHEADDOWNLO
STHIMGOODCOVERALRIGHTLISTENTHETERMINATORSANINFILTRATIONUNITPARTMANPARTMACHINEUNDERNEATHITSAHYPERALLLOYCOMBATCHASSISMICROPROCESSORCONTROLLEDPL
LYARMOREDVERYTOUGHBUTOUTSIDEITSLIVINGHUMANITISUEFLESHSKINHAIRBLOODGROWNFORTHECYBORGSLOOKRESEIKNOWYOUWANTTOHELPPUTPAYATTENTIONTHESERIESHADRUB
BERSKINWESPOTTEDTHEMEASYBUTTHESEARENEWTHEYLOOKHUMANSWEATBADBREATHEVERYTHINGVERYHARDTOSPOTIHADTOWAITTILHEMOVEDONYOUBEFOREICOULDZEROHIMHEYIMNOT
STUPIDYKNOWTHEYCANTBUILDDANYTHINGLIKE THATYETNONOYETNOTFORABOUTFORTYYEARSOITSFROMTHEFUTUREISTHATRIGHTONEPOSSIBLEFUTUREFOURYOURPOINTOFVIEWIDON
TKNOWTHETECHSTUFFANDYOU'REFROMTHEFUTURETOORIGHTRIGHTCYBORGSDONTFEELPAINIDODONTDO THATAGAINJUSTLETMEGOLISTENUNDERSTANDTHATTERMINATORISOUTTHEREIT
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Do you want to try another keyword? (yes/no):

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9. And I have the plaintext, so I will say no.
10. Upload your working code to Canvas as <netid>_lab03_part01.py

Lab 03 Template

- 1) Take a screenshot of the five most common trigrams. Make sure you copy all the data including counts, distances, and common divisor.
(10 total points, 2 points each trigram with full information)

- 2) **Why do we analyze trigrams (three-character sequences) when attempting to break a Vigenère cipher? What information do the differences between their positions provide?**
(10 total points, 5 points each question)
- 3) **How does calculating the greatest common divisor (GCD) of position differences help in determining the key length?**
(5 total points)
- 4) **Based on your findings, what do you suspect the key length is? Justify your answer.**
(10 total points, 5 points length value, 5 points justification)
- 5) **Why do you assume the most frequent character is E? How does it help us with multiple shift-by-n ciphers?**
(10 total points, 5 points each question)
- 6) **Take a screenshot for *each* shift-by-n cipher**
(10 total points)
- 7) **Decrypt the ciphertext using the potential key values you found in question 5. Show all iterations, the final keyword, and the final plaintext.**
(30 total points, 20 points for all iterations, 5 points for correct keyword, 5 points for correct plaintext)
- 8) **What strategies can you use to refine your keyword guesses if the plaintext is not immediately recognizable?**
(5 total points)
- 9) **Submit your documented python code as <netid>_lab03_part01.py to canvas.**
(10 total points for working code. No points awarded if the code is copy/pasted from someone else.)